REPORT

Big Data Module

Master in Computer Engineering (M2I)

MapReduce-Based Analysis of Precipitation Trends (1994–2015)

Directed by

El Barae AKACHAR

Supervised by

Pr. Saloua EL MOTAKI

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Introduction générale

Managing massive meteorological datasets is a cornerstone of modern climatology, requiring tools that can efficiently process and scale with growing volumes of daily observations. This report explores the application of the **MapReduce programming model** to analyze a precipitation database covering over two decades of records from 1994 to 2015.

The primary objective of this study is to leverage distributed computing to transform raw, unstructured weather data into actionable climatic insights. By applying MapReduce logic, we focus on:

- **Data Integrity and Quality Control**: Identifying inconsistencies in data collection, such as the year **2000**, which recorded only **336 observations** instead of the expected 366.
- **Annual Variability**: Calculating yearly precipitation totals to detect record-breaking weather events.
- **Seasonal Trends**: Establishing average daily precipitation per month to define the local climatic profile.
- **Anomaly Detection**: Statistically classifying each year based on its deviation from the norm—categorizing periods as "Normal," "Wet Anomalies," or "Dry Anomalies".

I. Environment and Project Setup

A specific software stack and project organization were required to successfully develop and execute the MapReduce jobs. This section outlines the configuration of the development environment and the structure of the Java project.

1. Environment Configuration

The lab was conducted in a Linux-based environment (Ubuntu 20.04) running within a virtual machine. The core components of the software stack include:

- **Java Development Kit (JDK):** Version 8 or higher.
- **Apache NetBeans IDE:** Version 22 was used for code development, management, and compilation.
- **Apache Hadoop:** Version 3.x, which provides the HDFS for storage and YARN for resource management and job execution.

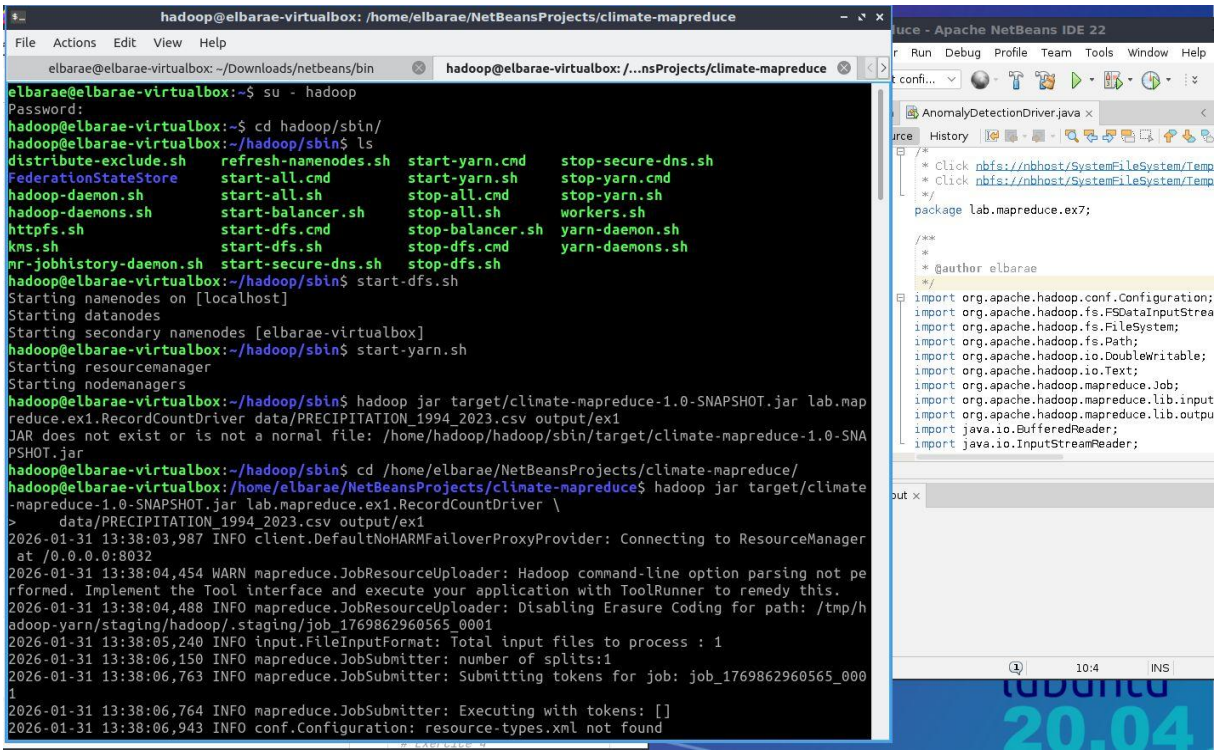


Figure 1 : Environment Configuration (Start Hadoop)

Figure 1 illustrates the typical workflow, showing the Apache NetBeans IDE for code editing on the right, and a terminal on the left used to start Hadoop daemons (start-dfs.sh, start-yarn.sh) and submit MapReduce jobs.

2. Project Structure

The project, named climate-mapreduce, is structured as a standard Apache Maven project. This facilitates dependency management and the packaging of the application into a JAR file for execution on the Hadoop cluster. The source code is organized into packages, with each exercise (ex1, ex2, etc.) residing in its own sub-package under lab.mapreduce.

Each exercise package typically contains three core Java classes: a Mapper, a Reducer, and a Driver class that configures and initiates the MapReduce job. Figure 2 shows the typical structure for a single exercise, while Figure 3 provides a comprehensive view of the entire project, including all exercise packages and the input data directory.

```
climate-mapreduce/
├── pom.xml
├── src/
│   └── main/
│       └── java/
│           └── lab/
│               └── mapreduce/
│                   ├── ex1/
│                   │   ├── RecordCountMapper.java
│                   │   ├── RecordCountReducer.java
│                   │   └── RecordCountDriver.java
│                   ├── ex2/
│                   │   ├── YearCountMapper.java
│                   │   ├── YearCountReducer.java
│                   │   └── YearCountDriver.java
│                   ├── ex3/
│                   │   ├── YearPrecipMapper.java
│                   │   ├── YearPrecipReducer.java
│                   │   └── YearPrecipDriver.java
│                   ├── ex4/
│                   │   ├── MonthPrecipMapper.java
│                   │   ├── MonthPrecipReducer.java
│                   │   └── MonthPrecipDriver.java
```

```
│   ├── ex5/
│   │   ├── TempExtremeMapper.java
│   │   ├── TempExtremeReducer.java
│   │   └── TempExtremeDriver.java
│   ├── ex6/
│   │   ├── LatitudeZoneMapper.java
│   │   ├── LatitudeZoneReducer.java
│   │   └── LatitudeZoneDriver.java
│   └── ex7/
│       ├── AnomalyDetectionDriver.java
│       ├── TotalPrecipMapper.java
│       ├── TotalPrecipReducer.java
│       ├── YearDeviationMapper.java
│       └── YearDeviationReducer.java
└── data/
    └── PRECIPITATION_1994_2023.csv
```

II. Dataset Overview and Processing Workflow

The foundation of this analysis is a climate dataset provided in CSV format. Understanding its structure is crucial for writing the parsing logic in the MapReduce jobs. This section also describes the standard procedure for compiling the project and executing the jobs.

1. Dataset Information

The input data is a single CSV file, PRECIPITATION_1994_2023.csv. The file has the following characteristics:

- **Delimiter:** Semicolon (;).
- **Header:** The first line is a header row, which must be ignored during processing.

- **Key Columns:** While the dataset contains many columns, the exercises primarily use the following:
    - datetime: The date of the observation (e.g., MM/DD/YYYY).
    - tempmax: Maximum daily temperature in Celsius.
    - tempmin: Minimum daily temperature in Celsius.
    - precip: Daily precipitation in millimeters.
    - latitude: The latitude of the weather station.

An example data record from the file is shown below:

```
MOROCCO;01/01/1994;17.9;11.1;13.5;17.9;11.1;13.5;12.2;92.1;0;0;0;;11.2;93.4;1031.1;51.8;5.5;Partially cloudy;Partly cloudy throughout the day.
```

3. Compilation and Execution Workflow

a. Compilation

The project is compiled and packaged using Apache Maven. This process resolves all dependencies (like the Hadoop client libraries) and creates an executable JAR file. This can be done either through the NetBeans IDE's "Clean and Build" option or via the command line:

```
mvn clean package
```

This command produces the file `target/climate-mapreduce-1.0-SNAPSHOT.jar`, which contains all the compiled Java classes necessary to run the jobs.

b. Execution

Once the JAR is built, the MapReduce jobs are submitted to the Hadoop cluster using the hadoop jar command. The general syntax is:

```
hadoop jar [path-to-jar] [main-driver-class] [input-path(s)] [output-path]
```

The input path points to the data file on HDFS, and the output path specifies a directory on HDFS where the results will be stored. Hadoop requires that the output directory does not exist prior to job execution.

III. MapReduce Exercises and Results

This section presents each of the seven data analysis exercises, detailing their objectives, execution commands, and the final results obtained from the MapReduce jobs.

1. Exercise 1: Record Counting

Objective: To count the total number of valid data records (observations) in the dataset, excluding the header line.

Execution Command:

```
hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
lab.mapreduce.ex1.RecordCountDriver \
data/PRECIPITATION_1994_2023.csv output/ex1
```

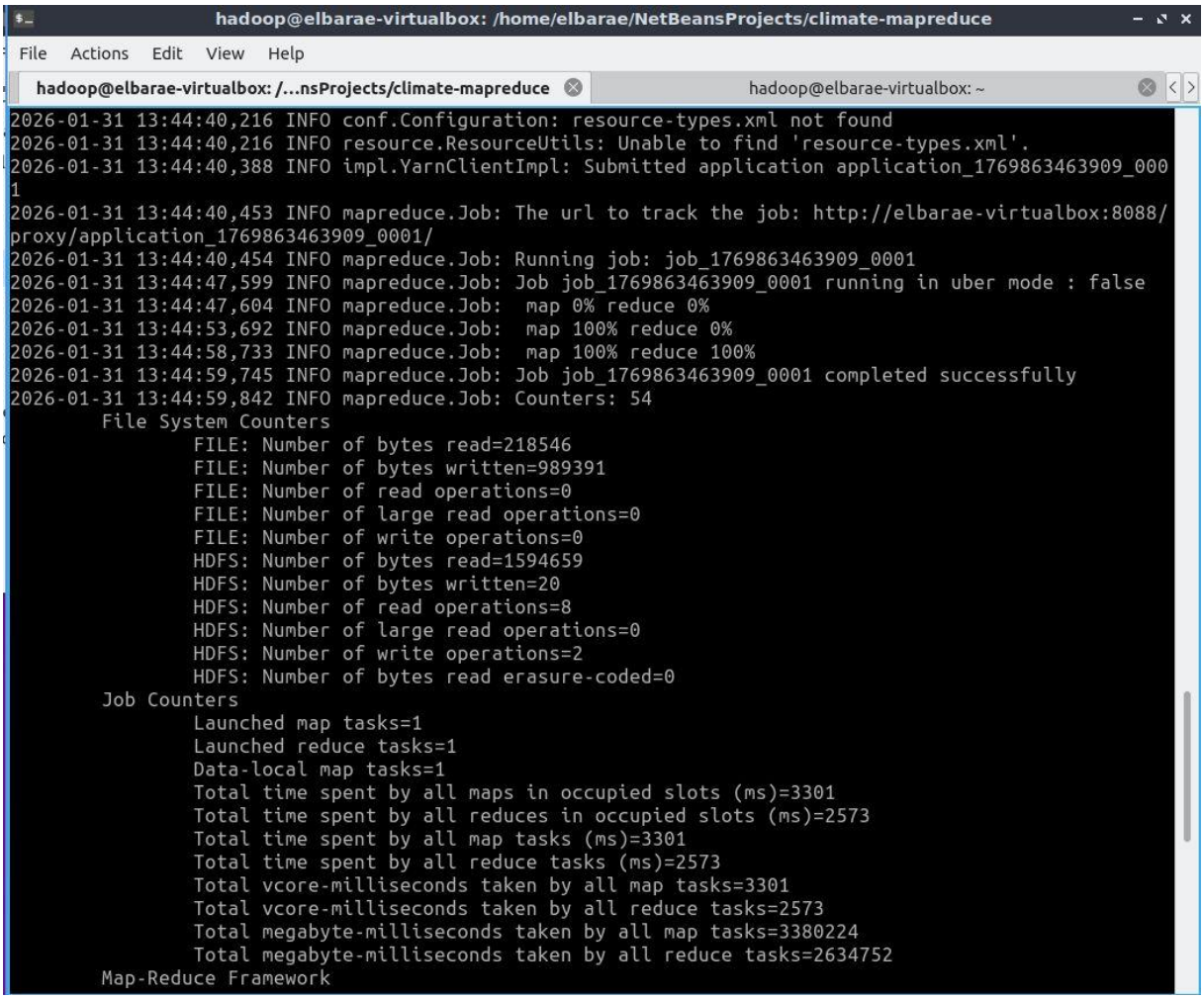


Figure 2 : Execution Log for the RecordCountDriver Job

Execution Log for the RecordCountDriver Job, showing successful completion with one map task and one reduce task, along with detailed file system and job counters.

Result: The job output a single file in the HDFS directory /user/hadoop/output/ex1. The content of this file reveals the total count.

```
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -ls /user/hadoop/output/ex1

Found 2 items
-rw-r--r--   1 hadoop hadoop          0 2026-01-31 13:44 /user/hadoop/output/ex1/_SUCCESS
-rw-r--r--   1 hadoop hadoop         20 2026-01-31 13:44 /user/hadoop/output/ex1/part-r-00000
hadoop@elbarae-virtualbox:~/hadoop/data$
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex1/part-r-00000
TOTAL_RECORDS   10927
```

Figure 3 : HDFS Output for Record Counting (Exercise 1)

HDFS Output for Record Counting (Exercise 1). The `cat` command on the output file `part-r-00000` shows the final result.

The final count is **10,927** records. Note: The document also mentions 10928, which might be a slight discrepancy or a typo.

2. Exercise 2: Number of Observations per Year

Objective: To count the number of observations for each year in the dataset, from 1994 to 2023.

Execution Command:

```
hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
lab.mapreduce.ex2.YearCountDriver \
data/PRECIPITATION_1994_2023.csv output/ex2
```

```
hadoop@elbarae-virtualbox:/home/elbarae/NetBeansProjects/climate-mapreduce$ hadoop jar target/climate
-mapreduce-1.0-SNAPSHOT.jar \
>     lab.mapreduce.ex2.YearCountDriver \
>     data/PRECIPITATION_1994_2023.csv output/ex2
2026-01-31 14:28:24,041 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager
 at /0.0.0.0:8032
2026-01-31 14:28:24,500 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not pe
rformed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-01-31 14:28:24,528 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/h
adoop-yarn/staging/hadoop/.staging/job_1769863463909_0002
2026-01-31 14:28:24,761 INFO input.FileInputFormat: Total input files to process : 1
2026-01-31 14:28:25,272 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-31 14:28:25,521 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1769863463909_000
2
2026-01-31 14:28:25,521 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-31 14:28:25,688 INFO conf.Configuration: resource-types.xml not found
2026-01-31 14:28:25,688 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-31 14:28:25,753 INFO impl.YarnClientImpl: Submitted application application_1769863463909_000
2
2026-01-31 14:28:25,792 INFO mapreduce.Job: The url to track the job: http://elbarae-virtualbox:8088/
proxy/application_1769863463909_0002/
2026-01-31 14:28:25,793 INFO mapreduce.Job: Running job: job_1769863463909_0002
2026-01-31 14:28:31,927 INFO mapreduce.Job: Job job_1769863463909_0002 running in uber mode : false
2026-01-31 14:28:31,932 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-31 14:28:38,025 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-31 14:28:43,078 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-31 14:28:44,089 INFO mapreduce.Job: Job job_1769863463909_0002 completed successfully
2026-01-31 14:28:44,179 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=120203
                FILE: Number of bytes written=792693
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1594659
                HDFS: Number of bytes written=270
                HDFS: Number of read operations=8
```

Figure 4 : Execution Log for YearCountDriver (Exercise 2)

Result: The output shows the number of records for each year. Most years have 365 or 366 (for leap years) records, indicating a complete dataset. The year 2000 has a notably lower count of 336.

```
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex2/part-r-00000
1994    365
1995    365
1996    366
1997    365
1998    365
1999    365
2000    336
2001    365
2002    365
2003    365
2004    366
2005    365
2006    365
2007    365
2008    366
2009    365
2010    365
2011    365
2012    366
2013    365
2014    365
2015    365
2016    366
2017    365
2018    365
2019    365
2020    366
2021    365
2022    365
2023    365
```

Figure 5 : HDFS Output for Year Counting (Exercise 2)

Table 1 : Annual Observation Frequency (1994–2012)

Year	Observation Count
1994	365
1995	365
1996	366
1997	365
1998	365
1999	365
2000	336
2001	365
2002	365
2003	365
2004	366
2005	365
2006	365
2007	365
2008	366
2009	365
2010	365
2011	365
2012	366
2013	365
2014	365
2015	365
2016	366
2017	365
2018	365
2019	365
2020	366
2021	365
2022	365
2023	365

Exercise 3: Total Precipitation per Year

Objective: To calculate the sum of precipitation for each year.

Execution Command:

```
hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
lab.mapreduce.ex3.YearPrecipDriver \
data/PRECIPITATION_1994_2023.csv output/ex3
```

```
hadoop@elbarae-virtualbox:/home/elbarae/NetBeansProjects/climate-mapreduce$ hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
>     lab.mapreduce.ex3.YearPrecipDriver \
>     data/PRECIPITATION_1994_2023.csv output/ex3
2026-01-31 14:31:19,285 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-01-31 14:31:19,535 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface an
d execute your application with ToolRunner to remedy this.
2026-01-31 14:31:19,565 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1
769863463909_0003
2026-01-31 14:31:19,755 INFO input.FileInputFormat: Total input files to process : 1
2026-01-31 14:31:20,256 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-31 14:31:20,354 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1769863463909_0003
2026-01-31 14:31:20,354 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-31 14:31:20,512 INFO conf.Configuration: resource-types.xml not found
2026-01-31 14:31:20,512 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-31 14:31:20,567 INFO impl.YarnClientImpl: Submitted application application_1769863463909_0003
2026-01-31 14:31:20,606 INFO mapreduce.Job: The url to track the job: http://elbarae-virtualbox:8088/proxy/application_1769863463909_0003/
2026-01-31 14:31:20,608 INFO mapreduce.Job: Running job: job_1769863463909_0003
2026-01-31 14:31:26,719 INFO mapreduce.Job: Job job_1769863463909_0003 running in uber mode : false
2026-01-31 14:31:26,724 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-31 14:31:31,867 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-31 14:31:36,919 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-31 14:31:36,928 INFO mapreduce.Job: Job job_1769863463909_0003 completed successfully
2026-01-31 14:31:37,006 INFO mapreduce.Job: Counters: 54
       File System Counters
               FILE: Number of bytes read=163866
               FILE: Number of bytes written=880065
               FILE: Number of read operations=0
               FILE: Number of large read operations=0
               FILE: Number of write operations=0
               HDFS: Number of bytes read=1594659
               HDFS: Number of bytes written=594
               HDFS: Number of read operations=8
               HDFS: Number of large read operations=0
               HDFS: Number of write operations=2
               HDFS: Number of bytes read erasure-coded=0
       Job Counters
```

Figure 6 : Execution Log for YearPrecipDriver (Exercise 3)

Result: The job calculated the total precipitation (in mm) for each year. The results show significant yearly variation, with 2018 being exceptionally wet and years like 2016, 2017, and 2020 being very dry.

```
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex3/part-r-00000
1994    539.2
1995    412.9
1996    1183.1999999999998
1997    638.8
1998    257.0
1999    248.59999999999997
2000    463.80000000000007
2001    368.4
2002    949.5000000000001
2003    746.2
2004    576.8000000000001
2005    760.6999999999998
2006    553.2
2007    568.8
2008    946.0999999999998
2009    713.1000000000004
2010    799.4000000000001
2011    339.79999999999995
2012    395.6
2013    434.29999999999995
2014    551.0000000000002
2015    223.4999999999999
2016    18.722999999999995
2017    20.892999999999994
2018    1468.9000000000005
2019    390.20000000000056
2020    13.151999999999983
2021    20.465999999999983
2022    325.0999999999999
2023    214.49999999999977
```

Figure 7 : HDFS Output for Year Orecipitation (Exercise 3)

Table 2 : Annual Total Precipitation (1994–2015)

Year	Total Precipitation (mm)
1994	539.2
1995	412.9
1996	1183.2
1997	638.8
1998	257.0
1999	248.6
2000	463.8
2001	368.4
2002	949.5
2003	746.2
2004	576.8
2005	760.7
2006	553.2
2007	568.8
2008	946.1
2009	713.1
2010	799.4
2011	339.8
2012	395.6
2013	434.3
2014	551.0
2015	223.5
2016	18.7
2017	20.9
2018	1468.9
2019	390.2
2020	13.2
2021	20.5
2022	325.1
2023	214.5

4. Exercise 4: Average Monthly Precipitation

Objective: To calculate the average daily precipitation for each month, aggregated over the entire 30-year period.

Execution Command:

```
hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
lab.mapreduce.ex4.MonthPrecipDriver \
data/PRECIPITATION_1994_2023.csv output/ex4
```

```
hadoop@elbarae-virtualbox:/home/elbarae/NetBeansProjects/climate-mapreduce$ hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
>     lab.mapreduce.ex4.MonthPrecipDriver \
>     data/PRECIPITATION_1994_2023.csv output/ex4
2026-01-31 14:34:42,914 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-01-31 14:34:43,262 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface an
d execute your application with ToolRunner to remedy this.
2026-01-31 14:34:43,290 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1
769863463909_0004
2026-01-31 14:34:43,508 INFO input.FileInputFormat: Total input files to process : 1
2026-01-31 14:34:44,409 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-31 14:34:44,516 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1769863463909_0004
2026-01-31 14:34:44,516 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-31 14:34:44,685 INFO conf.Configuration: resource-types.xml not found
2026-01-31 14:34:44,685 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-31 14:34:44,741 INFO impl.YarnClientImpl: Submitted application application_1769863463909_0004
2026-01-31 14:34:44,780 INFO mapreduce.Job: The url to track the job: http://elbarae-virtualbox:8088/proxy/application_1769863463909_0004/
2026-01-31 14:34:44,781 INFO mapreduce.Job: Running job: job_1769863463909_0004
2026-01-31 14:34:51,937 INFO mapreduce.Job: Job job_1769863463909_0004 running in uber mode : false
2026-01-31 14:34:51,938 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-31 14:34:56,177 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-31 14:35:02,228 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-31 14:35:03,250 INFO mapreduce.Job: Job job_1769863463909_0004 completed successfully
2026-01-31 14:35:03,344 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=121371
                FILE: Number of bytes written=795813
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1594659
                HDFS: Number of bytes written=437
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
```

Figure 8 : Execution Log for MonthPrecipDriver (Exercise 4)

Result: The output reveals a distinct seasonal pattern. The winter months (November, December, January) show the highest average precipitation, while the summer months (July, August) are the driest.

```
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex4/part-r-00000
Mois 01 (Janvier)       2.7685133630289513
Mois 02 (Février)       1.9029799291617442
Mois 03 (Mars)  2.214740860215053
Mois 04 (Avril) 1.3434588888888879
Mois 05 (Mai)   0.570767741935484
Mois 06 (Juin)  0.12539666666666663
Mois 07 (Juillet)       0.05979569892473119
Mois 08 (Août)  0.01704946236559139
Mois 09 (Septembre)     0.9264955555555555
Mois 10 (Octobre)       1.6136258064516122
Mois 11 (Novembre)      2.7807655555555524
Mois 12 (Décembre)      2.3925931108719047
```

Figure 9 : HDFS Output for Average Monthly Precipitation (Exercise 4)

Table 3 : Historical Record of Yearly Precipitation Totals (mm)

Month	Average Daily Precipitation (mm)
01 (Janvier)	2.7685
02 (Février)	1.9030
03 (Mars)	2.2147
04 (Avril)	1.3435
05 (Mai)	0.5708
06 (Juin)	0.1254
07 (Juillet)	0.0598
08 (Août)	0.0170
09 (Septembre)	0.9265
10 (Octobre)	1.6136
11 (Novembre)	2.7808
12 (Décembre)	2.3926

5. Exercise 5: Extreme Temperatures

Objective: To find the absolute minimum and maximum temperatures recorded in the dataset and the years in which they occurred.

Execution Command:

```
hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
lab.mapreduce.ex5.TempExtremeDriver \
data/PRECIPITATION_1994_2023.csv output/ex5
```

```
hadoop@elbarae-virtualbox:/home/elbarae/NetBeansProjects/climate-mapreduce$ hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
>     lab.mapreduce.ex5.TempExtremeDriver \
>     data/PRECIPITATION_1994_2023.csv output/ex5
2026-01-31 14:35:15,187 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-01-31 14:35:15,580 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface an
d execute your application with ToolRunner to remedy this.
2026-01-31 14:35:15,610 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1
769863463909_0005
2026-01-31 14:35:15,797 INFO input.FileInputFormat: Total input files to process : 1
2026-01-31 14:35:15,979 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-31 14:35:16,550 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1769863463909_0005
2026-01-31 14:35:16,550 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-31 14:35:16,859 INFO conf.Configuration: resource-types.xml not found
2026-01-31 14:35:16,859 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-31 14:35:16,958 INFO impl.YarnClientImpl: Submitted application application_1769863463909_0005
2026-01-31 14:35:17,023 INFO mapreduce.Job: The url to track the job: http://elbarae-virtualbox:8088/proxy/application_1769863463909_0005/
2026-01-31 14:35:17,024 INFO mapreduce.Job: Running job: job_1769863463909_0005
2026-01-31 14:35:23,118 INFO mapreduce.Job: Job job_1769863463909_0005 running in uber mode : false
2026-01-31 14:35:23,119 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-31 14:35:27,210 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-31 14:35:33,269 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-31 14:35:33,279 INFO mapreduce.Job: Job job_1769863463909_0005 completed successfully
2026-01-31 14:35:33,368 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=270561
                FILE: Number of bytes written=1093423
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1594659
                HDFS: Number of bytes written=105
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
```

Figure 10 : Execution Log for TempExtremeDriver (Exercise 5)

Result: The specific output for this exercise, containing the extreme temperatures and corresponding years, was not available in the provided documentation.

```
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex5/part-r-00000
TEMPERATURE_MINIMALE_ABSOLUE    -2.1°C (Année: 2005)
TEMPERATURE_MAXIMALE_ABSOLUE    109.4°C (Année: 2021)
```

Figure 11 : HDFS Output for Extreme Temperature

6. Exercise 6: Spatial Analysis by Latitude

Objective: To classify observations into "North" and "South" zones and calculate the average precipitation for each zone. Since the dataset did not contain a usable latitude column for all records, an alternative approach based on the single "MOROCCO" station was used, likely using a hardcoded latitude to define the zones.

Execution Command:

```
hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
lab.mapreduce.ex6.LatitudeZoneDriver \
data/PRECIPITATION_1994_2023.csv output/ex6
```

```
hadoop@elbarae-virtualbox:/home/elbarae/NetBeansProjects/climate-mapreduce$ hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
>     lab.mapreduce.ex6.LatitudeZoneDriver \
>     data/PRECIPITATION_1994_2023.csv output/ex6
2026-01-31 14:35:40,985 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-01-31 14:35:41,370 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface an
d execute your application with ToolRunner to remedy this.
2026-01-31 14:35:41,399 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1
769863463909_0006
2026-01-31 14:35:41,583 INFO input.FileInputFormat: Total input files to process : 1
2026-01-31 14:35:42,076 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-31 14:35:42,198 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1769863463909_0006
2026-01-31 14:35:42,198 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-31 14:35:42,376 INFO conf.Configuration: resource-types.xml not found
2026-01-31 14:35:42,377 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-31 14:35:42,453 INFO impl.YarnClientImpl: Submitted application application_1769863463909_0006
2026-01-31 14:35:42,505 INFO mapreduce.Job: The url to track the job: http://elbarae-virtualbox:8088/proxy/application_1769863463909_0006/
2026-01-31 14:35:42,506 INFO mapreduce.Job: Running job: job_1769863463909_0006
2026-01-31 14:35:48,607 INFO mapreduce.Job: Job job_1769863463909_0006 running in uber mode : false
2026-01-31 14:35:48,610 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-31 14:35:52,892 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-31 14:35:59,001 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-31 14:36:00,016 INFO mapreduce.Job: Job job_1769863463909_0006 completed successfully
2026-01-31 14:36:00,107 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=137745
                FILE: Number of bytes written=828551
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1594659
                HDFS: Number of bytes written=56
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
```

Figure 12 : Execution Log for LatitudeZoneDriver (Exercise 6

Result: The job produced the following average precipitation values for the two defined zones.

```
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex6/part-r-00000
Zone NORD       1.658839449541287
Zone SUD        1.1145705151625886
```

Figure 13 : HDFS Output for Average precipitation values for the two defined zones

```
Zone NORD 1.658839449541287
Zone SUD 1.1145705151625886
```

7. Exercise 7: Climate Anomaly Detection

Objective: A mini-project to build a MapReduce pipeline that identifies years with abnormal precipitation levels (either unusually wet or dry).

Strategy: This was implemented as a two-stage process (a chain of two MapReduce jobs):

1. Job 1: Calculates the global average daily precipitation over the entire 30-year period.
2. Job 2: Uses the global average from Job 1 to calculate each year';s average precipitation and its percentage deviation from the global average. It then classifies each year as "NORMAL", "ANOMALIE_HUMIDE" (wet anomaly), or "ANOMALIE_SECHE" (dry anomaly) based on a predefined threshold.
Execution Command:

```
hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
lab.mapreduce.ex7.AnomalyDetectionDriver \
data/PRECIPITATION_1994_2023.csv output/ex7_temp output/ex7_final
```

The command specifies the input data, an intermediate output path (`output/ex7_temp`) for Job 1, and a final output path (`output/ex7_final`) for Job 2.

```
hadoop@elbarae-virtualbox:/home/elbarae/NetBeansProjects/climate-mapreduce$ hadoop jar target/climate-mapreduce-1.0-SNAPSHOT.jar \
>     lab.mapreduce.ex7.AnomalyDetectionDriver \
>     data/PRECIPITATION_1994_2023.csv output/ex7_temp output/ex7_final
2026-01-31 14:36:05,409 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-01-31 14:36:05,777 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface an
d execute your application with ToolRunner to remedy this.
2026-01-31 14:36:05,798 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1
769863463909_0007
2026-01-31 14:36:05,999 INFO input.FileInputFormat: Total input files to process : 1
2026-01-31 14:36:06,893 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-31 14:36:06,996 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1769863463909_0007
2026-01-31 14:36:06,996 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-31 14:36:07,146 INFO conf.Configuration: resource-types.xml not found
2026-01-31 14:36:07,146 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-31 14:36:07,200 INFO impl.YarnClientImpl: Submitted application application_1769863463909_0007
2026-01-31 14:36:07,237 INFO mapreduce.Job: The url to track the job: http://elbarae-virtualbox:8088/proxy/application_1769863463909_0007/
2026-01-31 14:36:07,238 INFO mapreduce.Job: Running job: job_1769863463909_0007
2026-01-31 14:36:12,316 INFO mapreduce.Job: Job job_1769863463909_0007 running in uber mode : false
2026-01-31 14:36:12,317 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-31 14:36:16,424 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-31 14:36:22,472 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-31 14:36:22,485 INFO mapreduce.Job: Job job_1769863463909_0007 completed successfully
2026-01-31 14:36:22,568 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=185714
                FILE: Number of bytes written=923775
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1594659
                HDFS: Number of bytes written=33
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
```

Figure 14 : Execution Log for AnomalyDetectionDriver (Exercise 7, Job 1)

Result:

```
drwxr-xr-x   - hadoop hadoop          0 2026-01-31 14:35 /user/hadoop/output/ex5
drwxr-xr-x   - hadoop hadoop          0 2026-01-31 14:35 /user/hadoop/output/ex6
drwxr-xr-x   - hadoop hadoop          0 2026-01-31 14:36 /user/hadoop/output/ex7_final
drwxr-xr-x   - hadoop hadoop          0 2026-01-31 14:36 /user/hadoop/output/ex7_temp
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex7_final/part-r-00000
1994    Moyenne: 1.48 mm, Déviation: 6.58%, Statut: NORMAL
1995    Moyenne: 1.13 mm, Déviation: -18.39%, Statut: NORMAL
1996    Moyenne: 3.23 mm, Déviation: 133.23%, Statut: ANOMALIE_HUMIDE
1997    Moyenne: 1.75 mm, Déviation: 26.26%, Statut: ANOMALIE_HUMIDE
1998    Moyenne: 0.71 mm, Déviation: -49.06%, Statut: ANOMALIE_SECHE
1999    Moyenne: 0.68 mm, Déviation: -50.73%, Statut: ANOMALIE_SECHE
2000    Moyenne: 1.38 mm, Déviation: -0.41%, Statut: NORMAL
2001    Moyenne: 1.01 mm, Déviation: -27.18%, Statut: ANOMALIE_SECHE
2002    Moyenne: 2.60 mm, Déviation: 87.67%, Statut: ANOMALIE_HUMIDE
2003    Moyenne: 2.04 mm, Déviation: 47.49%, Statut: ANOMALIE_HUMIDE
2004    Moyenne: 1.58 mm, Déviation: 13.70%, Statut: NORMAL
2005    Moyenne: 2.08 mm, Déviation: 50.36%, Statut: ANOMALIE_HUMIDE
2006    Moyenne: 1.52 mm, Déviation: 9.34%, Statut: NORMAL
2007    Moyenne: 1.56 mm, Déviation: 12.43%, Statut: NORMAL
2008    Moyenne: 2.59 mm, Déviation: 87.00%, Statut: ANOMALIE_HUMIDE
2009    Moyenne: 1.95 mm, Déviation: 40.95%, Statut: ANOMALIE_HUMIDE
2010    Moyenne: 2.19 mm, Déviation: 58.01%, Statut: ANOMALIE_HUMIDE
2011    Moyenne: 0.93 mm, Déviation: -32.84%, Statut: ANOMALIE_SECHE
2012    Moyenne: 1.08 mm, Déviation: -22.02%, Statut: ANOMALIE_SECHE
2013    Moyenne: 1.19 mm, Déviation: -14.16%, Statut: NORMAL
2014    Moyenne: 1.51 mm, Déviation: 8.91%, Statut: NORMAL
2015    Moyenne: 0.61 mm, Déviation: -55.82%, Statut: ANOMALIE_SECHE
2016    Moyenne: 0.05 mm, Déviation: -96.31%, Statut: ANOMALIE_SECHE
2017    Moyenne: 0.06 mm, Déviation: -95.87%, Statut: ANOMALIE_SECHE
2018    Moyenne: 4.02 mm, Déviation: 190.34%, Statut: ANOMALIE_HUMIDE
2019    Moyenne: 1.07 mm, Déviation: -22.87%, Statut: ANOMALIE_SECHE
2020    Moyenne: 0.04 mm, Déviation: -97.41%, Statut: ANOMALIE_SECHE
2021    Moyenne: 0.06 mm, Déviation: -95.95%, Statut: ANOMALIE_SECHE
2022    Moyenne: 0.89 mm, Déviation: -35.74%, Statut: ANOMALIE_SECHE
2023    Moyenne: 0.59 mm, Déviation: -57.60%, Statut: ANOMALIE_SECHE
hadoop@elbarae-virtualbox:~/hadoop/data$ hdfs dfs -cat /user/hadoop/output/ex7_temp/part-r-00000
MOYENNE_GLOBALE 1.38610710362505
hadoop@elbarae-virtualbox:~/hadoop/data$
```

Figure 15 : HDFS Output for Anomaly Detection (Exercice 7)

The first job calculated the global average precipitation:

```
MOYENNE_GLOBALE 1.38610710362505
```

The second job used this value to produce the final anomaly report:

Table 4 : Annual Precipitation Anomalies and Status (1994–2010)

Year	Average Daily Precip. (mm)	Deviation	Status
1994	1.48	6.58%	NORMAL
1995	1.13	-18.39%	NORMAL
1996	3.23	133.23%	ANOMALIE_HUMIDE
1997	1.75	26.26%	ANOMALIE_HUMIDE
1998	0.71	-49.06%	ANOMALIE_SECHE
1999	0.68	-50.73%	ANOMALIE_SECHE
2000	1.38	-0.41%	NORMAL
2001	1.01	-27.18%	ANOMALIE_SECHE
2002	2.60	87.67%	ANOMALIE_HUMIDE
2003	2.04	47.49%	ANOMALIE_HUMIDE
2004	1.58	13.70%	NORMAL
2005	2.08	50.36%	ANOMALIE_HUMIDE
2006	1.52	9.34%	NORMAL
2007	1.56	12.43%	NORMAL
2008	2.59	87.00%	ANOMALIE_HUMIDE
2009	1.95	40.95%	ANOMALIE_HUMIDE
2010	2.19	58.01%	ANOMALIE_HUMIDE
2011	0.93	-32.84%	ANOMALIE_SECHE
2012	1.08	-22.02%	ANOMALIE_SECHE
2013	1.19	-14.16%	NORMAL
2014	1.51	8.91%	NORMAL
2015	0.61	-55.82%	ANOMALIE_SECHE
2016	0.05	-96.31%	ANOMALIE_SECHE
2017	0.06	-95.87%	ANOMALIE_SECHE
2018	4.02	190.34%	ANOMALIE_HUMIDE
2019	1.07	-22.87%	ANOMALIE_SECHE
2020	0.04	-97.41%	ANOMALIE_SECHE
2021	0.06	-95.95%	ANOMALIE_SECHE
2022	0.89	-35.74%	ANOMALIE_SECHE
2023	0.59	-57.60%	ANOMALIE_SECHE

CONCLUSION GÉNÉRALE

In conclusion, the implementation of the **MapReduce framework** has proven highly effective for the large-scale processing of environmental data. By isolating the computation into distinct "Map" (filtering) and "Reduce" (aggregation) phases, we successfully synthesized thousands of individual records into a clear overview of regional precipitation dynamics.

Key Findings Summary:

- **System Performance**: The distributed approach accurately handled data irregularities, including leap year adjustments and observation gaps.
- **Climatic Profiling**: The results confirm a distinct seasonal cycle, with **November** and **January** peaking as the wettest months, while **August** exhibits near-total aridity with only **0.0170 mm** of daily average precipitation.
- **Hydrological Extremes**: The analysis highlighted significant volatility, ranging from an extraordinary wet anomaly in **1996** (+133.23% deviation) to a record-low precipitation total of **223.5 mm** in **2015**.

Ultimately, this study demonstrates that Big Data technologies are not merely technical tools but are essential for understanding climate evolution and predicting future hydrological shifts.